

Question	Answer	Marks
6(a)	<u>sum of</u> current(s) into junction = <u>sum of</u> current(s) out of junction or (algebraic) sum of current(s) at a junction is zero	B1
6(b)(i)	$R = V / I$	C1
	$= 0.60 / 7.5 \times 10^{-3}$	C1
	$= 80 \Omega$	A1
6(b)(ii)	resistance decreases	B1
6(c)(i)	$E = 0.60 + 0.30$	A1
	$= 0.90 \text{ V}$	
6(c)(ii)	$(I =) 9.3 - 7.5$	C1
	$I = 1.8 \text{ (mA) or } 1.8 \times 10^{-3} \text{ (A)}$	A1
	$R = 0.90 / 1.8 \times 10^{-3}$	
	$= 500 \Omega$	
	or	
	total resistance $= 0.90 / 9.3 \times 10^{-3} = 96.8 \text{ (}\Omega\text{)}$ total resistance of diode and X $= 0.90 / 7.5 \times 10^{-3} = 120 \text{ (}\Omega\text{)}$ $1 / 96.8 = 1 / R + 1 / 120$	(C1)
	$R = 500 \Omega$	(A1)